

These notes are based on the two uploaded PPTs. Repeated material is consolidated, while the deeper OSI/TCP-IP material from Module-1-2 is retained.

0. 1-HOUR REVISION PLAN

- 0–10 min: Computer-network basics, components, PAN/LAN/MAN/WAN, topologies, client-server vs P2P.
- 10–25 min: OSI 7 layers — memorize order + one main function + examples/PDU.
- 25–35 min: Layered architecture, peer-to-peer process, encapsulation and OSI layer responsibilities.
- 35–45 min: TCP/IP model + OSI comparison + IP/ARP/RARP/ICMP/IGMP/TCP/UDP/SCTP.
- 45–55 min: Internet structure, IPv4/IPv6, DNS, network protocols.
- 55–60 min: Network applications + cloud/virtualization networking + last-minute comparisons.

1. INTRODUCTION TO COMPUTER NETWORKS

1.1 What is a Computer Network?

- A collection of interconnected computing devices (computers, servers, mobile devices) that exchange data and share resources using a common communication medium and set of rules.
- Devices are called nodes or hosts.
- Connections can be wired (cable, fibre) or wireless (radio, Wi-Fi).
- Enables sharing of data, hardware and applications.
- Communication is governed by protocols — agreed rules for formatting, sending and receiving data.

INTRODUCTION

What Is a Computer Network?

A collection of interconnected computing devices (computers, servers, mobile devices) that can exchange data and share resources using a common communication medium and set of rules.

- Devices are called nodes or hosts
- Connected via wired links (cable, fibre) or wireless links (radio, Wi-Fi)
- Enables sharing of data, hardware and applications
- Governed by a set of rules called protocols



Diagram from Module-1: interconnected network devices.

1.2 Why Computer Networks Matter

- Resource Sharing — share printers, storage, software and Internet access.
- Communication — email, messaging and video calls connect people instantly.
- Centralised Data — central storage/management improves consistency and backup.
- Scalability & Cost — users/services can be added incrementally and hardware duplication reduced.

1.3 Core Components of a Network

Component	Meaning
Node / Host	Any device that sends or receives data — computers, servers, phones, printers.
Link / Medium	Physical or wireless path for data — cable, fibre optic, radio waves.
Switch / Router	Intermediate devices that direct data to the correct destination.
NIC	Network Interface Card; hardware connecting a device to the network medium.
Protocol	Agreed rules defining how data is formatted, sent and received.

1.4 Types of Networks by Geographic Scale

Feature	PAN	LAN	MAN	WAN
Full name	Personal Area Network	Local Area Network	Metropolitan Area Network	Wide Area Network
Range	~10 m	Building / campus; under 1 km in comparison table	5–50 km / city	Country / global
Primary use	Personal gadgets	Single building	Whole city	Continents / global
Data speed	Moderate (up to 24 Mbps)	Very high (up to 10 Gbps)	High (up to 1 Gbps)	Lower; varies by region
Technology	Bluetooth, Zigbee	Ethernet, Wi-Fi	Fiber-optic, Microwave	Leased lines, Satellite
Ownership	Single individual	Single private organization	Joint / telecom company	Multiple organizations
Setup cost	Very low	Low–moderate	High	Very high
Error rate	Very low	Low	Moderate	High

INTRODUCTION

Types of Networks by Geographic Scale

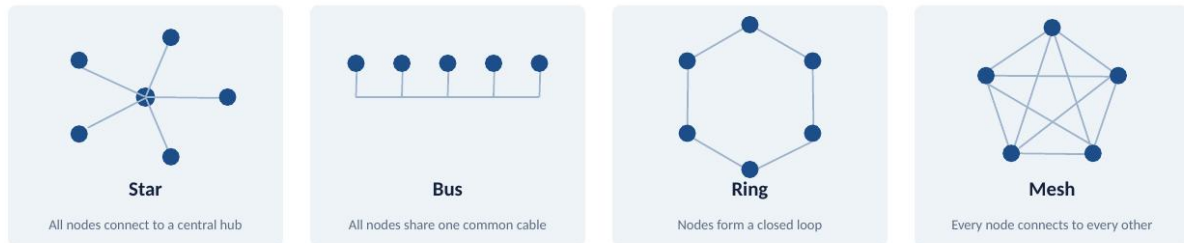


PAN, LAN, MAN and WAN visual from Module-1.

1.5 Network Topologies

Topology	Basic structure / definition
Star	All nodes connect to a central hub.
Bus	All nodes share one common cable.
Ring	Nodes form a closed loop.
Mesh	Every node connects to every other node.

Network Topologies



Star, Bus, Ring and Mesh diagrams from Module-1.

1.6 Client–Server vs Peer-to-Peer (P2P)

Feature	Client–Server	Peer-to-Peer (P2P)
Structure	Dedicated server provides resources/services.	Every peer can act as both client and server.
Communication	Clients request data/services from server.	Peers share resources directly.
Control	Centralised control; easier security & backup.	No central point of control; more fault-tolerant.
Examples	Web servers, email servers, databases.	File-sharing networks, blockchain nodes.

Client–Server vs Peer-to-Peer (P2P)



Client–Server vs P2P visual from Module-1.

2. NETWORK MODELS & LAYERED ARCHITECTURE

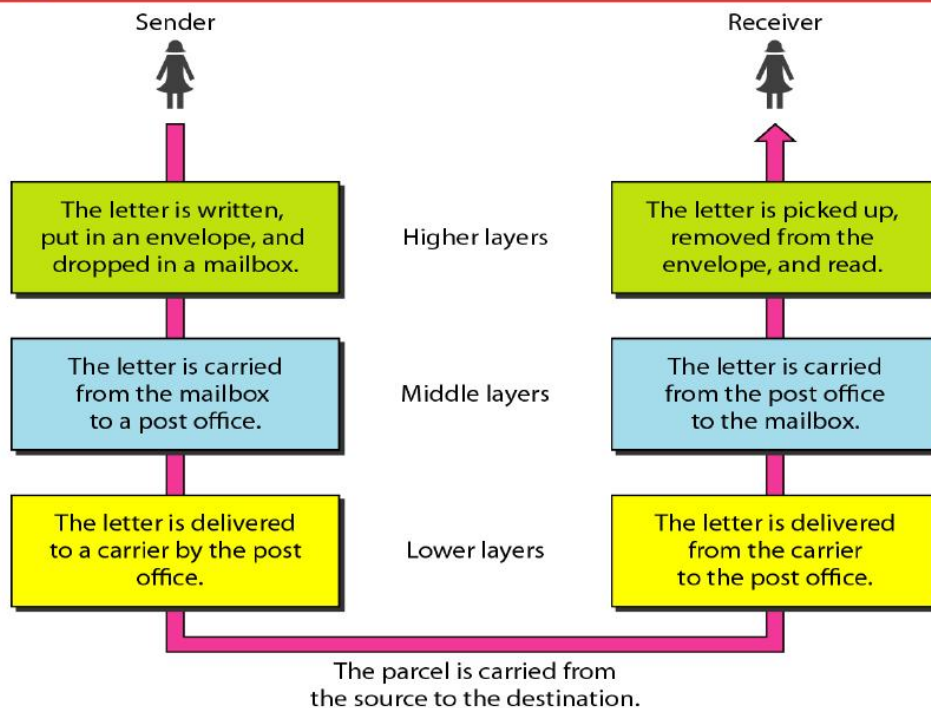
2.1 Why Use Layered Network Models?

- Network communication is broken into independent layers, each handling one specific job.
- Modularity — each layer can be designed/updated independently.

- Simplifies design — complex networking tasks become smaller, manageable problems.
- Interoperability — standard layers allow devices from different vendors to communicate.
- Easier troubleshooting — faults can be isolated to a particular layer.
- Each layer provides a specific set of services through protocols and uses services from adjacent layers.
- Layering decomposes the network-design problem and provides a modular design; changing one layer need not affect the whole system.

Module-1-2 explains network architecture as general blueprints guiding network design/implementation, e.g. OSI architecture and Internet architecture; abstraction of services/procedures naturally leads to layering.

Figure 2.1 Tasks involved in sending a letter



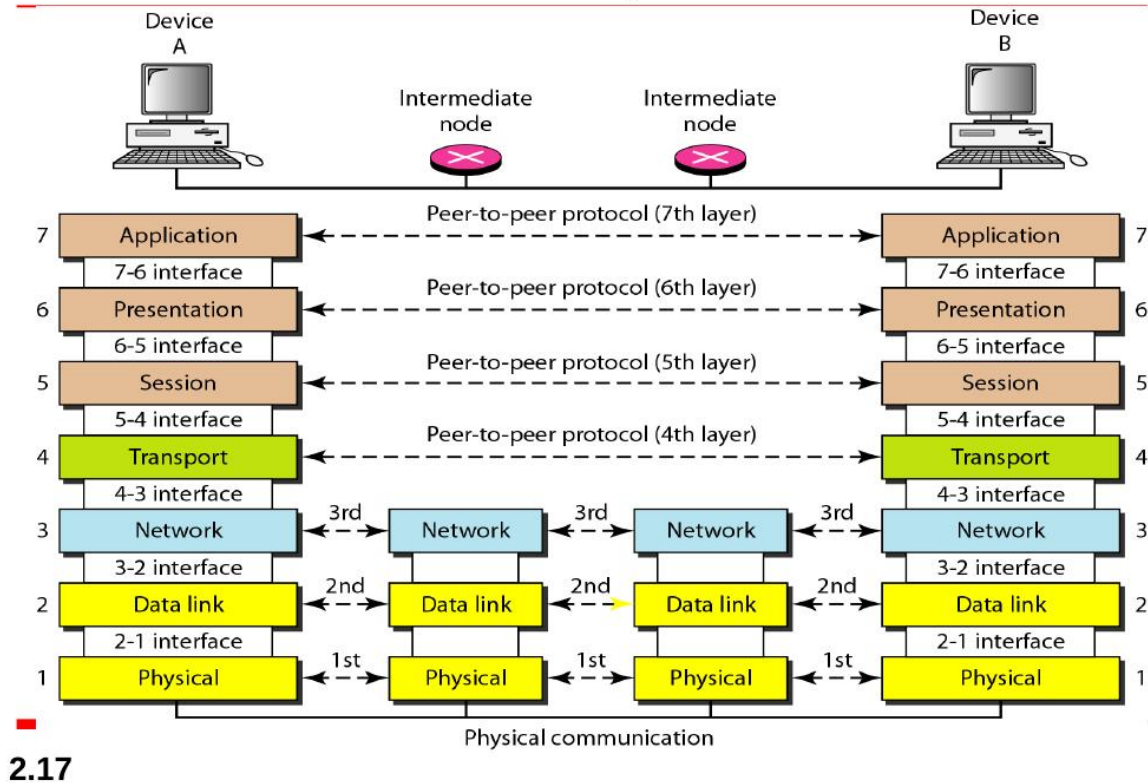
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Layered-task analogy from the detailed Network Models PPT: sending a letter through postal services.

2.2 Hierarchy and Layer Interaction

- Tasks must be performed in the order specified by the hierarchy of layers.
- Within one machine, a layer calls services of the layer below it and provides services to the layer above it.
- Between machines, layer x on one machine communicates logically with layer x on the other using protocols.
- Communication between machines is a peer-to-peer process at the corresponding layer.
- Interfaces between adjacent layers allow data/network information to pass down during sending and up during receiving.

The interaction between layers in the OSI model



Interaction between layers in the OSI model.

2.3 Main Benefits of Layering

- Breaks a complex networking problem into manageable components.
- Provides modularity and well-defined functions/interfaces.
- Allows individual layer functionality to be modified without redesigning the whole system.
- Supports interoperability and easier troubleshooting.

3. OSI MODEL

3.1 Definition

- OSI = Open Systems Interconnection.
- ISO = International Standards Organization; OSI is the model.
- OSI is a layered framework for network communication between different computer systems.
- It has seven separate but related layers, each defining part of the process of moving information across a network.
- It is a conceptual/reference model rather than a model implemented directly.

The OSI Model — 7 Layers

7	Application	Interfaces directly with end-user software
6	Presentation	Data translation, encryption, compression
5	Session	Establishes & manages sessions between hosts
4	Transport	Reliable end-to-end delivery (TCP/UDP)
3	Network	Logical addressing & routing (IP)
2	Data Link	Physical addressing, framing (MAC)
1	Physical	Transmission of raw bits over medium

Memory Aid

"All People Seem To Need Data Processing"

Layers 7 → 1: Application, Presentation, Session, Transport, Network, Data Link, Physical

Developed by ISO as a conceptual reference model — it is not implemented directly, but used to understand and design network communication.

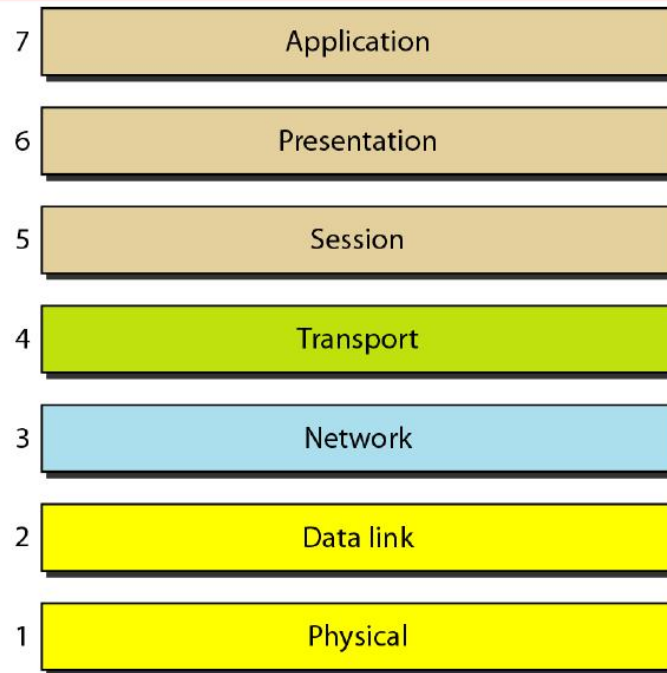
OSI 7-layer summary from Module-1.

Memory aid from Module-1: "All People Seem To Need Data Processing" = Application, Presentation, Session, Transport, Network, Data Link, Physical (7 → 1).

3.2 OSI Layers — Functions, Examples and PDU

Layer	Main function	Examples / PDU
7 Application	User-facing services and network access.	HTTP, FTP, SMTP; Data
6 Presentation	Data translation/format, encryption, compression.	SSL/TLS, JPEG, ASCII; Data
5 Session	Opens, manages and closes sessions; dialog/synchronization.	NetBIOS, RPC; Data
4 Transport	End-to-end delivery; reliable or unreliable transport.	TCP, UDP; Segments
3 Network	Logical addressing and path selection/routing.	IP, ICMP; Packets
2 Data Link	Node-to-node delivery, framing, physical addressing/error detection.	Ethernet, MAC; Frames
1 Physical	Transmission of raw bits over physical medium.	Cables, hubs; Bits

Figure 2.2 *Seven layers of the OSI model*



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Seven OSI layers from the detailed Network Models PPT.

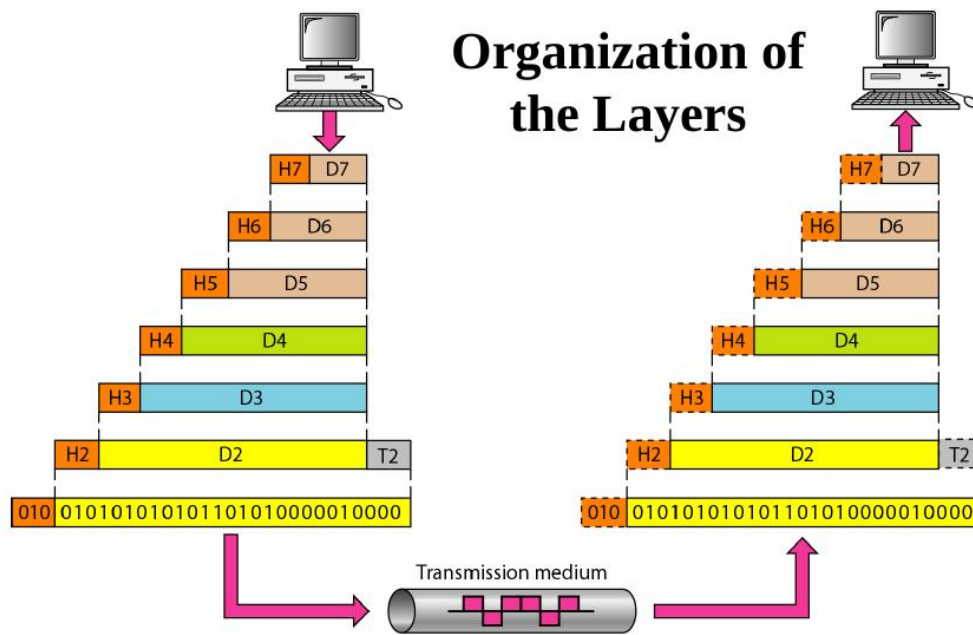
3.3 OSI Layer Responsibilities — Detailed

- Physical Layer: carries bit stream over physical medium; handles mechanical/electrical interface and transmission-medium specifications.
- Physical also covers: physical characteristics of interfaces/medium, bit representation, data rate, bit synchronization, line configuration, physical topology and transmission mode.
- Data Link Layer: transforms physical layer into a reliable link and makes physical layer appear error-free to the network layer.
- Data Link provides hop-to-hop delivery; responsibilities include framing, physical addressing, flow control, error control and access control.
- Network Layer: delivers individual packets from source host to destination host; handles logical addressing and routing.
- Network layer is especially needed when systems are on different networks connected by intermediate devices.
- Transport Layer: delivers a message from one process to another; unlike network-layer source-to-destination delivery, transport is process-to-process delivery of the entire message.
- Transport responsibilities: service-point addressing, segmentation/reassembly, connection control, flow control and error control.
- Session Layer: dialog control and synchronization.
- Presentation Layer: syntax/semantics of exchanged information; translation, encryption and compression.
- Application Layer: enables users/software to access the network; supports e-mail, remote file access/transfer, shared databases and distributed information services.
- Application responsibilities: network virtual terminal; file transfer/access/management; mail services; directory services.

3.4 OSI Layer Organization

- Layers 1–3 (Physical, Data Link, Network) = network support layers.
- Layers 5–7 (Session, Presentation, Application) = user support layers; allow interoperability among unrelated software systems.
- Layer 4 (Transport) links the two groups and ensures lower-layer output is usable by upper layers.
- Upper OSI layers are almost entirely software; lower layers combine software/hardware, while Physical is mostly hardware.

Figure 2.4 *An exchange using the OSI model*



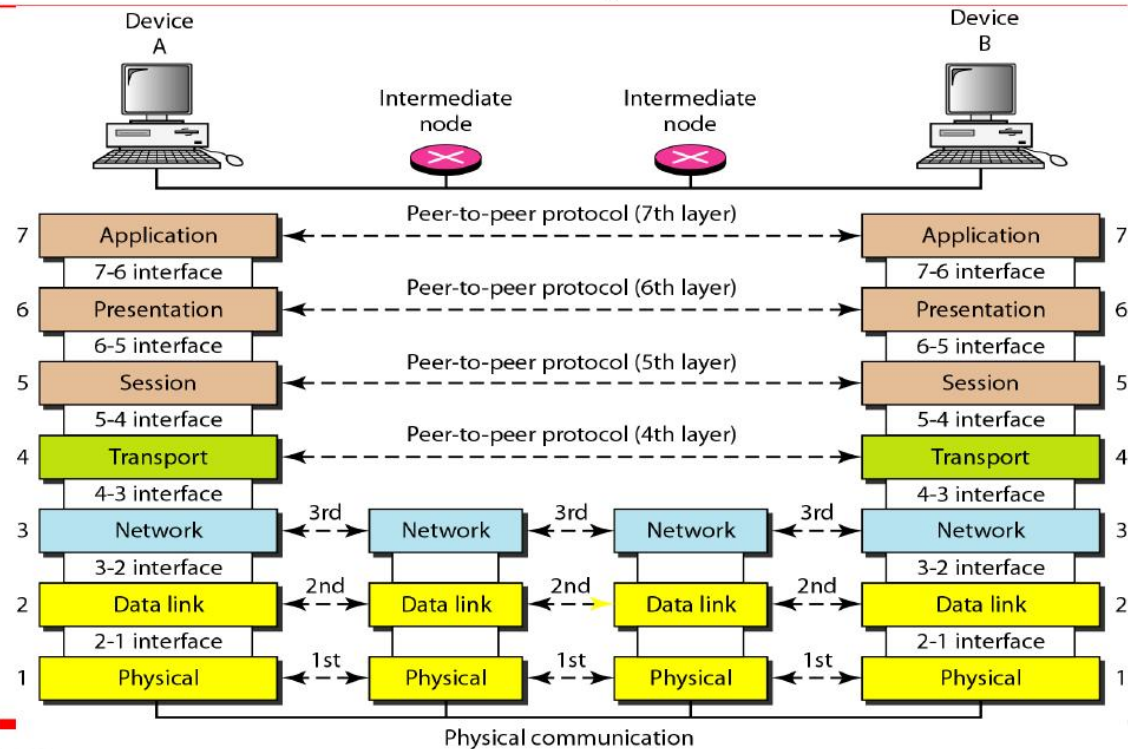
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OSI exchange/organization and encapsulation-style visual from the detailed PPT.

3.5 Peer-to-Peer Process in OSI

- At the Physical layer, device A directly sends a stream of bits to device B after converting bits into a transmittable form.
- At higher layers, information moves down the layers on device A, across to device B, then back up the layers.
- Corresponding layers communicate logically using their layer-specific protocols.
- Adjacent-layer interfaces control the passing of data and network information.

The interaction between layers in the OSI model



2.17

Use this diagram to explain peer-to-peer layer interaction.

3.6 Encapsulation

- Encapsulation is a key aspect of OSI data communication.
- At level N-1, the data portion carries the whole packet received from level N, including its data, header and possibly trailer.
- Level N-1 does not need to know which part is data versus header/trailer; it treats the complete incoming packet as one logical unit.
- In simple exam language: as data moves down the stack, each layer adds its control information; at the receiver, the process is reversed.

Encapsulation

- Another aspect of data communications in the OSI model is **encapsulation**.
 - The data portion of a packet at level **N-1** carries the whole packet (data and header and maybe trailer) from level **N**.
 - Level **N-1** is not aware of which part of the encapsulated packet is data and which part is the header or trailer.
 - For level N-1, the whole packet coming from level N is treated as one logical unit.

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Encapsulation concept from the detailed Network Models PPT.

4. TCP/IP PROTOCOL SUITE

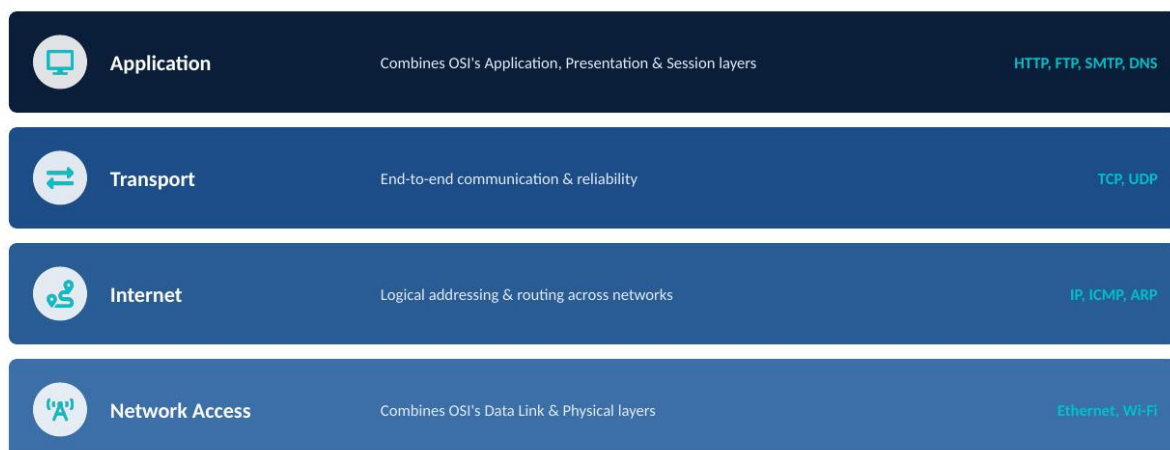
4.1 TCP/IP Model

- Module-1 presents a practical 4-layer TCP/IP model used by the modern Internet, developed by the US DoD before OSI.
- Application — combines OSI Application + Presentation + Session; examples: HTTP, FTP, SMTP, DNS.
- Transport — end-to-end communication and reliability; TCP, UDP.
- Internet — logical addressing and routing; IP, ICMP, ARP.
- Network Access — combines OSI Data Link + Physical; Ethernet, Wi-Fi.

NETWORK MODELS

The TCP/IP Model — 4 Layers

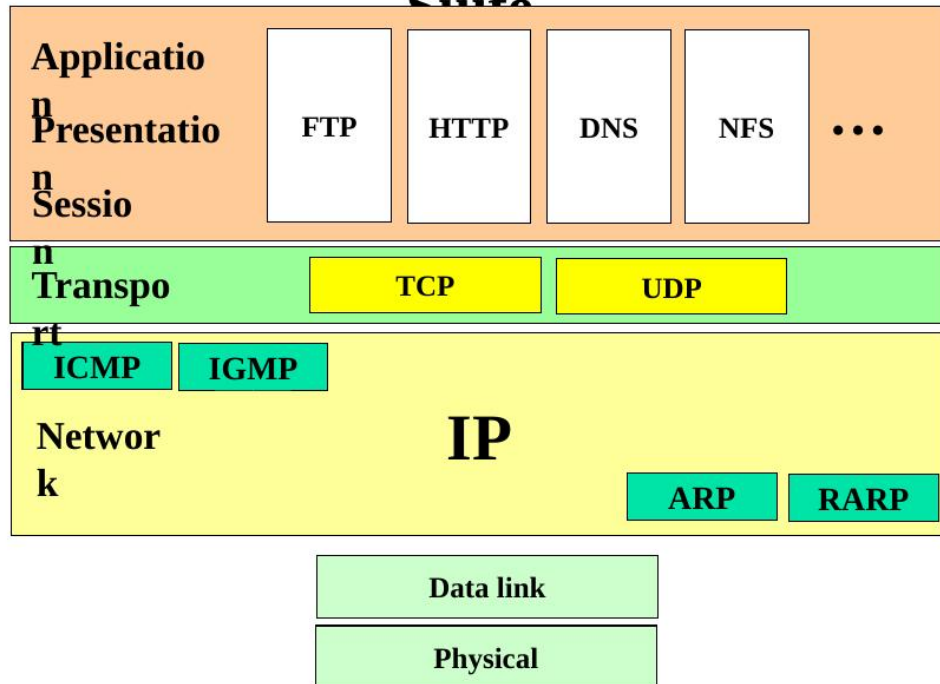
A practical, implementation-based model that the modern Internet is actually built on — developed by the US DoD before OSI.



4.2 Important Nuance from the Detailed TCP/IP PPT

- The detailed PPT states that the original TCP/IP protocol suite was defined with four layers: host-to-network, Internet, transport and application.
- When compared to OSI, the detailed PPT also presents a 5-layer view: Physical, Data Link, Network, Transport and Application.
- Therefore, in an exam, follow the layer version used by the question/PPT: the Module-1 practical model is 4 layers, while the detailed PPT explicitly discusses a 5-layer comparison to OSI.

The Layers in the TCP/IP Protocol Suite



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Figure 2.16 *TCP/IP and OSI model*

TCP/IP and OSI relationship shown in the detailed PPT.

4.3 TCP/IP is Hierarchical

- TCP/IP is a hierarchical protocol made of interacting modules, each providing specific functionality.
- Its layers contain relatively independent protocols.
- Hierarchical means each upper-level protocol is supported by one or more lower-level protocols.

4.4 TCP/IP Layer Details

- Physical + Data Link: no single specific protocol is defined; TCP/IP supports standard and proprietary protocols, including LAN and WAN technologies.
- Network / Internetwork layer: supports IP plus ARP, RARP, ICMP and IGMP.
- Transport layer: responsible for message delivery from one process to another.
- Application layer: equivalent to the combined Session, Presentation and Application layers of OSI.

4.5 Key TCP/IP Network-Layer Protocols

Protocol	What to write
IP	Most important TCP/IP network-stack protocol; implements internetworking; unreliable, connectionless, best-effort delivery; host-to-host protocol.
ARP	Finds the physical/NIC address when the network address is known.
RARP	Finds the Internet/network address when the physical address is known.
ICMP	Hosts/gateways use it to send notification of packet/datagram problems back to the sender.
IGMP	Facilitates simultaneous transmission of messages to a group of recipients.

4.6 Key TCP/IP Transport Protocols

- UDP — adds port addresses, checksum error control and length information to upper-layer data; connectionless and fast.
- TCP — reliable and connection-oriented.
- SCTP/STCP — supports newer applications such as voice over the Internet; combines best features of UDP and TCP according to the PPT.

PROTOCOLS

TCP vs UDP — Transport Layer Comparison

	TCP	UDP
Connection	Connection-oriented (handshake)	Connectionless
Reliability	Guaranteed delivery, error checking	No delivery guarantee
Ordering	Data arrives in order	Order not guaranteed
Speed	Slower due to overhead	Faster, minimal overhead
Use Cases	Web, email, file transfer	Video streaming, gaming, VoIP

TCP vs UDP comparison from Module-1.

5. THE INTERNET

5.1 What is the Internet and How is it Structured?

- The Internet is a vast global “network of networks” connecting billions of devices using the standardised TCP/IP protocol suite.
- ISPs (Internet Service Providers) connect homes/organisations to the Internet backbone.
- Backbone networks use high-capacity fibre links to carry data between major regions.
- Routers and exchange points direct traffic efficiently between interconnected networks.
- Typical path in the PPT: You → Local ISP → Internet Backbone → Destination ISP → Web Server.

What Is the Internet & How Is It Structured?

The **Internet** is a vast, global “network of networks” that connects billions of devices worldwide using the standardised TCP/IP protocol suite.



ISPs (Internet Service Providers)

Provide access by connecting homes/organisations to the Internet backbone



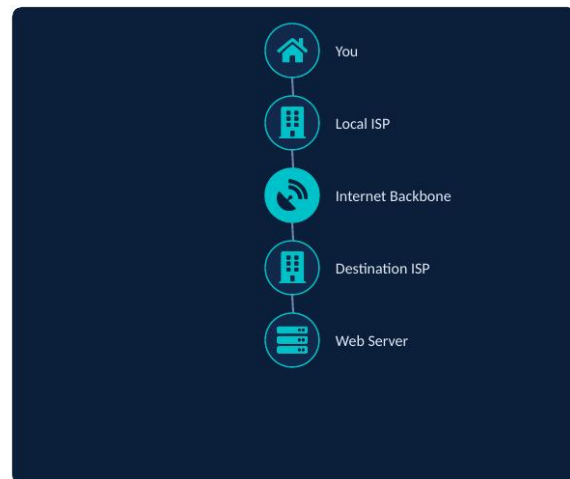
Backbone Networks

High-capacity fibre links that carry data between major regions globally



Routers & Exchange Points

Direct traffic efficiently between thousands of interconnected networks



Internet structure/path diagram from Module-1.

5.2 IP Addressing — IPv4 vs IPv6

- Every device on a network needs a unique IP address to send and receive data.

Feature	IPv4	IPv6
Address size	32-bit	128-bit
Example	192.168.1.1	2001:0db8::1
Address count	~4.3 billion	3.4×10^{38}
Notation	4 decimal numbers, each 0–255	8 groups of hexadecimal
Main issue/goal	Address exhaustion due to device growth	Designed to solve IPv4 exhaustion

IP Addressing — IPv4 vs IPv6

Every device on a network needs a unique IP address to send and receive data — like a postal address for the digital world.

IPv4

32-bit

192.168.1.1

- ~4.3 billion unique addresses
- Written as 4 decimal numbers (0-255)
- Address exhaustion due to device growth

IPv6

128-bit

2001:0db8::1

- 3.4×10^{38} unique addresses
- Written in 8 groups of hexadecimal
- Designed to solve IPv4 exhaustion

IPv4 vs IPv6 visual from Module-1.

5.3 DNS — Domain Name System

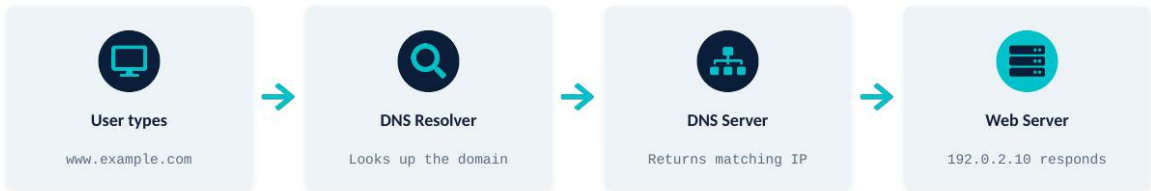
- DNS translates human-friendly domain names into the numeric IP addresses used by computers.
- Often called the “phonebook of the Internet.”

- Flow: user enters `www.example.com` → DNS resolver looks up domain → DNS server returns matching IP → web server at that IP responds.

THE INTERNET

Domain Name System (DNS)

DNS translates human-friendly domain names into the numeric IP addresses computers use to identify each other — often called the “phonebook of the Internet.”



DNS resolution flow from Module-1.

6. NETWORK PROTOCOLS

6.1 What is a Network Protocol?

- A protocol is a formal set of rules and conventions governing how devices communicate, format, transmit and interpret data over a network.
- Syntax — structure/format of data; how bits are organised into fields.
- Semantics — meaning of each field; what action/response it indicates.
- Timing — when data should be sent and at what speed it should arrive.

6.2 Key Network Protocols

Protocol	Function
HTTP / HTTPS	Transfers web pages; HTTPS adds encryption.
FTP	Transfers files between client and server.
SMTP	Sends email between mail servers.
DNS	Resolves domain names to IP addresses.
TCP	Reliable, connection-oriented data delivery.
UDP	Fast, connectionless data delivery.
IP	Addresses and routes packets across networks.
SSH	Secure remote login and command execution.

6.3 TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented; handshake	Connectionless
Reliability	Guaranteed delivery + error checking	No delivery guarantee
Ordering	Data arrives in order	Order not guaranteed
Speed	Slower due to overhead	Faster; minimal overhead
Use cases	Web, email, file transfer	Video streaming, gaming, VoIP

7. NETWORK APPLICATIONS

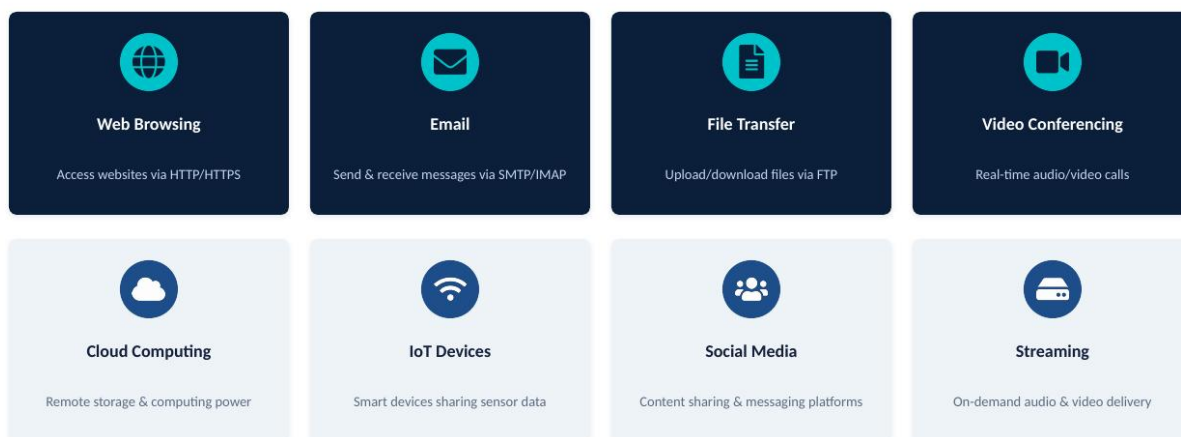
7.1 Everyday Applications

- Web Browsing — access websites using HTTP/HTTPS.

- Email — send/receive messages using SMTP/IMAP as listed in Module-1.
- File Transfer — upload/download using FTP.
- Video Conferencing — real-time audio/video calls.
- Cloud Computing — remote storage and computing power.
- IoT Devices — smart devices share sensor data.
- Social Media — content sharing and messaging.
- Streaming — on-demand audio/video delivery.

NETWORK APPLICATIONS

Everyday Applications Built on Networks



Everyday network applications from Module-1.

7.2 Why Networking Matters for Virtualization & Cloud

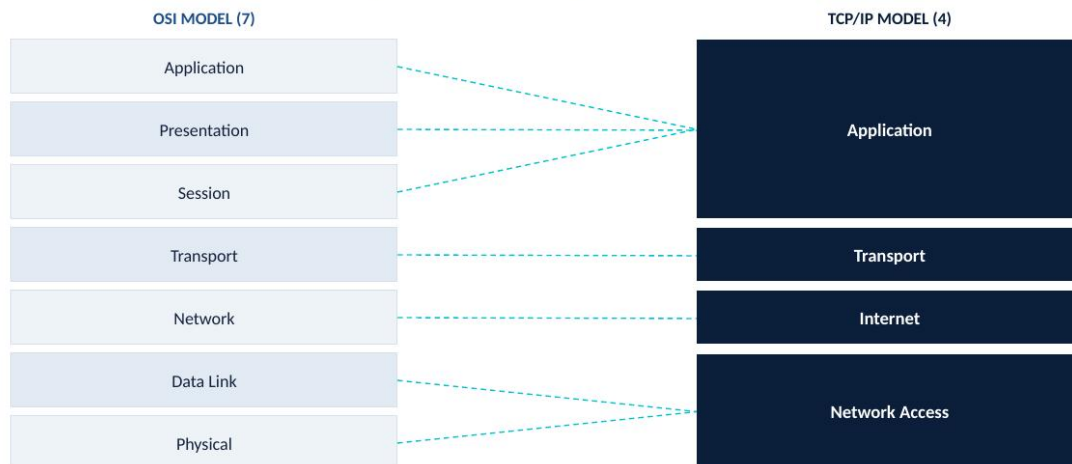
- Virtual Machine Connectivity — virtual switches and NICs allow VMs on the same host or across hosts to communicate.
- Cloud Service Delivery — IaaS, PaaS and SaaS are delivered to users over the network.
- Distributed & Shared Storage — SAN/NAS and cloud storage are accessed remotely by virtualized workloads.
- Virtual Networks & VLANs — software-defined networking isolates/connects VMs across data centres.
- Load Balancing & Scaling — traffic is routed across VM/container instances to scale cloud applications.
- APIs for Cloud Management — provisioning and orchestration tools such as hypervisor/cloud APIs run over HTTP.

8. EXAM HIGH-YIELD COMPARISONS & MEMORY

8.1 OSI vs TCP/IP

Point	OSI	TCP/IP
Layers	7	4 in Module-1 practical model; detailed PPT also discusses a 5-layer comparison
Nature	Conceptual/reference model	Practical implementation-oriented Internet protocol suite
Upper layers	Application, Presentation, Session separate	Combined into Application
Lower layers	Data Link + Physical separate	Combined as Network Access in 4-layer view
Network layer	Network	Internet
Transport	Transport	Transport

OSI vs TCP/IP — Side-by-Side Comparison



OSI vs TCP/IP side-by-side mapping from Module-1.

8.2 Delivery Types to Remember

- Data Link = hop-to-hop delivery.
- Network = source-to-destination / host-to-host packet delivery.
- Transport = process-to-process delivery of the message.
- Physical = actual bit transmission over the medium.

8.3 One-Line Layer Memory

- Physical → bits / medium / signals.
- Data Link → frames / MAC / hop-to-hop / error & flow control.
- Network → packets / IP / logical addressing / routing.
- Transport → segments / TCP-UDP / process-to-process / reliability.
- Session → dialog control / synchronization.
- Presentation → translation / encryption / compression.
- Application → user/network services / email / file transfer / directory services.

8.4 Last-Minute Definitions

Computer Network — Interconnected devices that exchange data/share resources through communication media and protocols.

Protocol — Formal rules governing communication, including syntax, semantics and timing.

OSI — Seven-layer conceptual reference model for network communication.

TCP/IP — Hierarchical Internet protocol suite; practical model for Internet communication.

Encapsulation — Each lower layer treats the higher-layer packet as its data and adds its own control information.

IP — Connectionless, unreliable, best-effort host-to-host internetworking protocol.

ARP — Maps known network address to physical/NIC address.

DNS — Maps human-friendly domain names to IP addresses.

TCP — Reliable, connection-oriented transport protocol.

UDP — Fast, connectionless transport protocol.

SCTP/STCP — Transport protocol described in the PPT as combining features of TCP and UDP and supporting newer applications such as Internet voice.

9. FINAL 60-SECOND CHECKLIST

- Know PAN/LAN/MAN/WAN ranges, use cases, technology, ownership, cost and error-rate differences.
- Draw/recognise Star, Bus, Ring and Mesh.
- Know Client–Server vs P2P.
- Memorize all 7 OSI layers in order.
- Know one main function + one example/PDU for every OSI layer.
- Know network support layers (1–3), user support layers (5–7), and Transport as the bridge.
- Know peer-to-peer process and encapsulation.
- Know TCP/IP 4-layer model and the detailed PPT's 5-layer comparison note.
- Know IP, ARP, RARP, ICMP, IGMP, TCP, UDP and SCTP/STCP.
- Know Internet structure, IPv4 vs IPv6 and DNS flow.
- Know protocol syntax, semantics and timing.
- Know TCP vs UDP.
- Know everyday network applications and how networking supports cloud/virtualization.

Exam answer pattern: Definition → 3–5 precise points → example/diagram where applicable → comparison table if asked. For diagrams, label layers clearly and write one function beside each major layer.